

Design of a 6-DoF Magnetorheological Fluid Based Parallel Haptic Robot for Surgical Teleoperation

Abstract—Haptic robots play a vital role in medical teleoperation systems by providing surgeons with accurate force feedback. Magnetorheological fluid (MRF)-based haptic devices have several advantages over conventional systems, especially in fast force generation and backdrivability. In this study, we developed a 6-DoF haptic robot based on MRF to provide high-performance force feedback. This MRF robot has a parallel structure that enhances its force generation stiffness and stability. With an innovative driving module configuration, the robot can be driven by a single motor to generate forces in six DoFs. To address the difficulty of forward kinematics derivation for the parallel robot, we combine the iteration optimization and space vector methods. To overcome the challenge associated with Jacobian calculation, we employ structure decomposition and reintegration for screw theory derivation. Compared with existing MRF-based devices, the proposed design enables active force and torque generation in all six DoFs using a single-motor clutch-based architecture while maintaining low reflected inertia at the operator interface. Experimental results demonstrate the feasibility of the proposed architecture and its proof-of-concept application in benchtop surgical teleoperation. Comprehensive characterization of Z-width, force-rendering bandwidth, maximum stable stiffness, and force/torque resolution remains for future work.

Index Terms—Haptic robot, teleoperation, magnetorheological fluid, parallel mechanism.

I. INTRODUCTION

HAPTIC interfaces are used in teleoperation to convey force information from a remote robot to its human operator. They also measure the operator's motion commands and transmit them to the remote robot, enabling bidirectional interaction. This bidirectional interaction enables intuitive remote manipulation and increases operational safety in many inaccessible areas, like remote surgery [1]. Transparency and stability are two fundamental indices for evaluating the teleoperation [2]. High transparency indicates prompt and accurate feedback to the operator. Stability determines the practical usability of the robot in all kinds of conditions.

The haptic robots in which forces are directly generated by electric motors, such as the Sigma.7 and Haption systems [3], are widely deployed in medical and industrial applications. However, motor-based force generation robots suffer from persistent challenges. Motor-driven haptic devices are widely used because they can provide high free-space transparency, relatively large workspaces, and accurately controlled active forces. Their contact stability, however, depends on the combined effects of device dynamics, controller design, sampling rate, communication delay, and environmental stiffness. MRF-based actuation provides a complementary approach by combining actively controlled force generation with passive energy dissipation through fluid shear.

MRF has attracted a lot of attention in recent years for its low operating inertia and fast force generation. It is a kind of smart fluid that can adjust its shear stress according to the magnetic flux density [4]. In the early years, MRF was used for vibration absorption, such as automotive isolators and seismic dampers [5]. Recently, the shear working mode of MRF has shown advantages in sophisticated stress generation and low motion friction. As a result, with the rotating disc configuration, MRF is gradually being applied in some tiny and delicate force generation devices [6]. Chen et al. introduced a rotary dual clutch using current to generate magnetic field for MRF stress modulation. They analysed the creep phenomenon and developed a fractional-order model to describe it [7]. Liu et al. studied the nonlinearity of the MRF clutch torque and proposed the modified Bouc-Wen model to describe the hysteresis. With parameter identification, this model maintained accurate predictions even in frequency response tests. However, the designed robot has limited DoFs and cannot deal with complex tasks [8]. Despite progress in miniaturisation and force generation modeling, there are still barriers to the prevalence of MRF devices in general teleoperation. The passivity and limited DoFs are two major factors that hinder the MRF devices from handling complex tasks.

The natural passivity of MRF stems from its resistance to motion. This characteristic weakens transparency during interaction due to the lack of active force generation. To compensate for this, the motor is usually integrated into the MRF system. Chapuis et al. proposed a hybrid MRF actuator by taking the motor into account for active force generation in haptic applications. To increase the mechanical bandwidth of the actuator, they introduced redundant clutch actuation, combining one motor and two clutches in opposite rotation directions. This endows the device active force generation with a short delay [9]. Similarly, Kikuchi et al. presented a dual motor actuator with compact multi-layer disc clutches, containing a pair of MRF clutches in opposite rotation directions. Experiments verified its fast bidirectional force generation capability [10]. Despite the active force generation capability of the MRF actuator, the multiple DoFs requirement for practical surgical tasks still remains unsolved. Najmaei et al. developed a 2-DoF haptic MRF robot with only one motor through a delicate pulley configuration. This prototype increased the system DoF and promoted its practical usability to some extent [11]. However, this MRF robot cannot maintain compact volume and low inertia in pursuing active force generation.

In addition to studies on active force generation, some research has focused on multiple DoF MRF robots for practical surgical tasks. Jong-Seok et al. introduced a 4-DoF MRF

haptic robot with two bidirectional MRF clutch pairs and two unidirectional MRF clutches. This device was designed for robot-assisted minimally invasive tele-surgery and could provide active force feedback in two directions [12]. A robot with 5 bidirectional clutch pairs has been designed by Pisetskiy et al. and demonstrated successfully as a serial manipulator. But its serial architecture inherently leads to high operating inertia and it needs further improvement for practical use [13]. Based on the above work, Cha et al. developed a 6-DoF MR-based haptic robot with decoupled rotation and translation modules. This robot could generate force in six directions and follow the dexterous hand postures of the operator. Although this design endows the robot active force generation in translation directions, no active torque feedback is achieved in rotation directions, degrading the force perception fidelity in teleoperation [14], [15].

Retrospecting the development of MRF robots for telesurgery, the fast force regeneration and backdrivability have been emphasized, which are crucial to telesurgery transparency. Whereas, there are still inescapable challenges in the MRF robot's practical application. The most obvious is the insufficient active force generation. Until now, no full active force generation in 6 DoFs is realized in MRF robots. This hinders the immersion and transparency of the telesurgery. Besides, the lack of integration with appropriate control method and hardware also weakens the low inertia feature of MRF devices, deteriorating the operating fidelity.

The MRF based robot holds promising potential for telesurgery. First, surgical requirements, like low inertia, contact stability, inherent safety, align with MRF advantages. Second, the workspace requirements of the targeted benchtop surgical tasks are compatible with the relatively small workspace of the proposed robot, although the limited workspace remains a constraint for tasks requiring larger translational or rotational motion.

In this study, the aforementioned limitations were fully considered during the design of the MRF haptic robot. It possesses 6 DoFs with active force generation and a compact driving module, implemented by a well-suited parallel mechanism. The utilisation of pulley transmission enables the driving of all MRF clutches using a single motor, thereby promoting the integration, while maintaining low operating inertia. The optimization based forward kinematics, in conjunction with screw theory based Jacobian derivation, enable the parallel MRF haptic robot to demonstrate active six-DoF force and torque generation and inherent damping stability during contact. For control, we adopt a model predictive control (MPC) framework adapted from [16]. The choice of MPC is motivated by its ability to handle constraints and predict future system states, which is particularly beneficial for MRF clutches with their inherent nonlinearities and fast dynamics. This study focuses on benchtop validation and proof-of-concept evaluation of the proposed device architecture. The proposed robot represents a promising step toward delivering surgical haptic feedback. Animal and cadaveric experiments are reserved for future work.

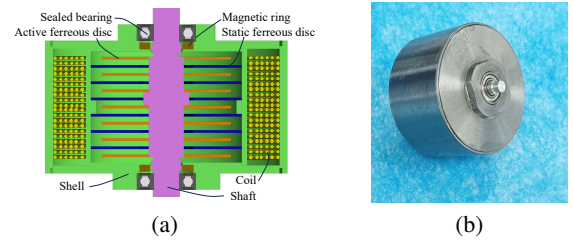


Fig. 1. (a) The section viewing of the MRF clutch. (b) Picture of the MRF clutch.

II. MRF AND MRF CLUTCH

The MRF is a type of suspension containing magnetizable ferrous particles and a carrier fluid [17], [18]. It has a key application in haptic devices, due to its capacity to modify shear strength in response to applied magnetic field. The field-dependent rheological response of MRF can occur on a millisecond time scale. However, the end-to-end response of an MRF-based haptic system also depends on sensing, electromagnetic actuation, control, computation, communication, and mechanical dynamics.

According to the stress property of MRF, we designed a rotational MRF clutch as illustrated in Fig. 1. Referring to an up-to-date study on continuously activated MRF seals [19], we use a permanent magnetic ring to seal the clutch against leakage, as shown in Fig. 1a. This method reduces friction by establishing a magnetic field, avoiding physical contact. This approach addresses the leakage and environmental contamination concerns associated with conventional MRF device seals. This MRF clutch comprises static and rotational discs, with gaps between them replenished by MRF. By proper distribution, the coil generates magnetic field virtually across the gaps, through the MRF. The magnetic flux intensity is determined by the applied current amplitude in the coil. Thus, the interactive force between these two kinds of discs can be regulated through the current. By connecting the static and rotational discs to the reference and the operator respectively, this MRF clutch can apply specific force to the operator.

III. MECHANICAL DESIGN

A. Overall Structure

The overall structure of the 6-DoF haptic robot is displayed in Fig. 2. It comprises three parts, the mobile platform, the driving system and the kinematic chains that link the

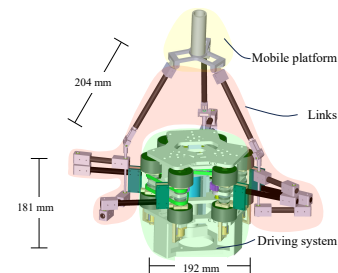


Fig. 2. Overview and constitution of the haptic robot.

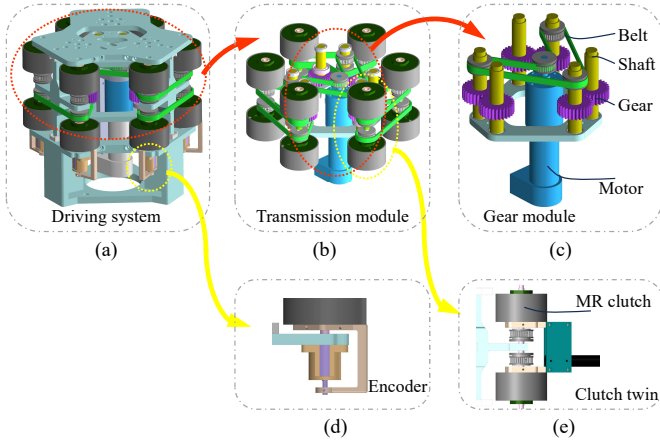


Fig. 3. Hardware decomposition of the driving system in the haptic robot.

former two parts. The mobile platform is the upper part, and an operating handle is located at its geometric center. The operator holds the handle and the robot interacts with operator's hand by regenerating force feedback. The varying pose of the handle presents the operator's hand pose, reflecting the operator's intention. The driving system is located at the bottom and acts as the force generation source. This part is fixed to the base, keeping the system stationary during operations. The kinematic chains connect the mobile platform with the driving system. Three kinematic chains are distributed around the driving system and concatenate the mobile platform circumferentially.

B. Driving System Design

The hierarchical constitutional diagram of the driving system is presented in Fig. 3. In designing this robot, we employed multiple MRF clutches. Conventional design approaches often use multiple motors with serial mechanical design for active force generation, weakening the MRF device's low inertia feature. Conversely, our innovative driving configuration with parallel structure enables active force generation of all clutches using a single motor. As illustrated in Fig. 3c, our proposed configuration is a compact, multiple-gear module with only one motor. Leveraging synchronous belts and reversing gears enables this module to feature six shafts, arranged in three pairs, with each pair comprising one clockwise and one anticlockwise shaft. The overall arrangement of the transmission module consists of 12 MRF clutches that form six clutch twins. Each clutch twin comprises two clutches in a top-bottom configuration, as shown in the Fig. 3e. The clutch twins are dispersed along the shafts and driven via synchronous belts. Notably, the rotational directions of the two clutches within each clutch twin are opposite. In operation, the top and bottom MRF clutch central shafts are driven rotating along opposite directions at constant velocity. Since the torque of each clutch can be adjusted separately using different current commands, the comprehensive torque output of each clutch twin is the integration of these separated opposite torques. Therefore, the clutch twin is equipped with the capacity of continuous, active, and dual-direction torque

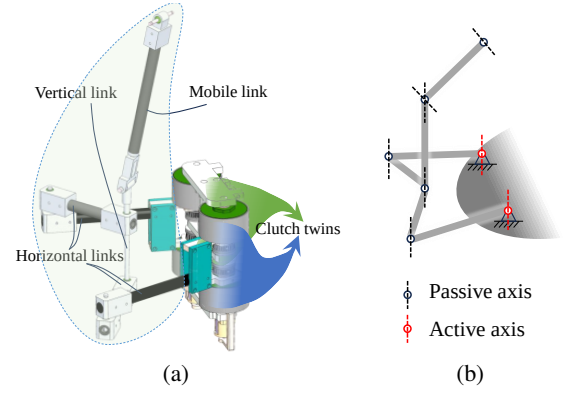


Fig. 4. (a) One kinematic chain in the haptic robot. (b) Simplified motion schematic diagram of the kinematic chain.

output. With proper belts and gears, all these six clutch twins have the capacity to generate continuous dual-direction torques. In summary, these clutch twins possess the capability of active force generation using one motor.

As a critical part, the motor with sufficient output is adopted (EC-4pole 22, Maxon Group, Switzerland). By coupling proper planetary reducer (GP32A, Maxon Group, Switzerland) and driver (EPOS4-70/15, Maxon Group, Switzerland), the output torque of this motor can reach up to 4.5 Nm, adequate for haptic feedback, mechanical friction, and efficiency decline. Miniature Hall-effect sensors are used for rotational measurement (QY1503-ANA5E, ACCNT, China). Besides the MRF clutches and the motor system, other supporting parts are made of Poly lactic acid.

To our knowledge, this is the first MRF clutch driving system capable of generating active forces in six dimensions, especially with just one motor. This development forms the basis for the active force regeneration of MRF robots. This architecture enables active bidirectional force and torque generation in all six DoFs and provides the basis for further development of six-DoF MRF-based haptic interfaces.

C. Kinematic Chain Motion Analysis

As illustrated in Fig. 4a, three kinematic chains with identical configurations are utilised to combine the mobile platform and the driving system. The links are made of carbon fiber rod for its lightweight and high stiffness. Each kinematic chain is connected to two adjacent clutch-twins at the bottom terminals and the mobile platform at the top terminal. The simplified motion structure of each kinematic chain is shown in Fig. 4b. Each kinematic chain has six links and eight rotational axes. Two of these axes are endowed with active force and driven by clutch twins. Accordingly, the motion of the four interconnected links between two adjacent clutch twins is limited to the horizontal plane. Thus, the vertical link in Fig. 4b can move horizontally, but remains vertical to horizontal plane. The top link is designed to connect the vertical link to the mobile platform.

IV. KINEMATIC ANALYSIS AND CONTROL

For the practical implementation of this MRF haptic robot, it must possess two fundamental functions. Firstly, the haptic

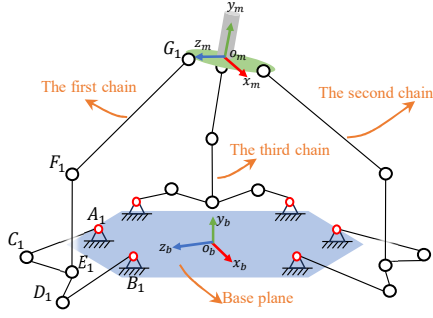


Fig. 5. Simplified joints and links of the haptic robot.

robot must be able to perceive the operator's hand pose [20]. As encoders are installed with the haptic robot's clutch-twins, forward kinematics are necessary to calculate the handle pose. Secondly, the haptic robot must apply the desired force to the handle [21]. This requires the derivation of the transformation mapping from the clutch-twin outputs to the handle.

A. Forward Kinematics

Unlike serial robots which can acquire forward kinematics directly from DH parameters and the angles of joints, the calculation of parallel robot forward kinematics can be challenging in the case of structure closed-loop constraints [22]. The encoders are installed at the base of this MRF parallel robot, resulting in many joint angles imperceptible in the kinematic chains. Consequently, there is an absence of comprehensive kinematic chain angles, rendering the DH method inapplicable [23].

Alternatively, the iterative optimization method is often used to recognise unknown angles based on geometric relations [24]. In this study, the iteration method and space vector consistency are combined to deduce these inaccessible joint values and the forward kinematics.

1) *Coordinate Frame Definition:* For clarity, the left kinematic chain is designated as the first chain. The second and third kinematic chains are arranged around the central axis of the driving system in an anticlockwise direction, as shown in Fig. 5. As illustrated, the joints are labelled with letters and the subscript indicates the specific chain. This parallel robot has two coordinate frames. The first is the base frame, labelled $o_b x_b y_b z_b$ located at the centre of the driving system in the horizontal base plane. The $o_b y_b$ axis is perpendicular to the plane and the $o_b z_b$ axis points to the first chain. The second frame is the mobile platform (handle) frame $o_m x_m y_m z_m$ with origin o_m located at the platform centre. The $o_m y_m$ axis is perpendicular to the platform plane and points upwards, with the $o_m z_m$ axis points to the connecting joint of the first chain.

The positional and length parameters regarding the first chain are listed in Table. I. Parameters regarding the other two chains can be obtained through analogous definition.

The pose of the handle in base frame $\mathbf{T}_m^b \in SE(3)$ is expressed as

$$\mathbf{T}_m^b = \begin{bmatrix} \mathbf{R}_m^b & \mathbf{o}_m^b \\ \mathbf{0} & 1 \end{bmatrix} \quad (1)$$

TABLE I
SYMBOL DEFINITION AND VALUES

Position / Length	Symbol	Value (mm)
A_1 in base frame	$A_1^b = (A_{1x}^b, A_{1y}^b, A_{1z}^b)$	(-39, 0, 68)
B_1 in base frame	$B_1^b = (B_{1x}^b, B_{1y}^b, B_{1z}^b)$	(39, 0, 68)
G_1 in platform frame	$G_1^m = (G_{1x}^m, G_{1y}^m, G_{1z}^m)$	(0, 0, 50)
Horizontal links	l_{AC}, l_{BD}	196
Horizontal links	l_{CE}, l_{DE}	166
Vertical link	l_{EF}	40
Mobile link	l_{FG}	204

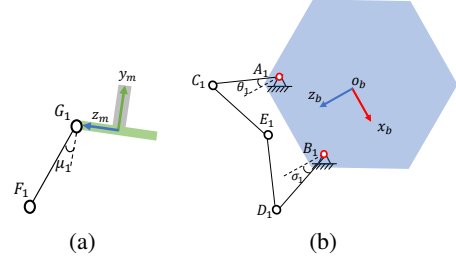


Fig. 6. (a) The schematic angle of the joint on the mobile platform. (b) The schematic angles of joints on the base plane.

where matrix $\mathbf{R}_m^b \in SO(3)$ is the direction cosine matrix indicating the orientation, $\mathbf{o}_m^b \in \mathbb{R}^3$ represents the position shift from o_b to o_m expressed in base frame.

2) *Numeric Iteration based Joint Recognition:* Taking the first chain for example, position of G_1 expressed in mobile platform frame is calculated as

$$\mathbf{G}_1^m = [G_{1x}^m \ G_{1y}^m \ G_{1z}^m]^T \quad (2)$$

Since G_1 is fixed to the platform, $\mathbf{G}_1^m$ is constant. Referring to Fig. 6a, denote the angle between link F_1G_1 and platform normal as μ_1 . Then, the position of F_1 in platform frame is

$$\mathbf{F}_1^m = \mathbf{G}_1^m + \mathbf{R}_x(\mu_1) [0 \quad -l_{FG} \quad 0]^T \quad (3)$$

where $\mathbf{R}_i(\lambda)$ denotes the direction cosine matrix of a rotation transformation around axis $i (i \in \{x, y, z\})$ for angle λ .

From another perspective, the position of F_1 can also be expressed in the base frame. Since the encoders are installed along the MRF clutch twin axes, the joint angles at A_1 and B_1 can be obtained directly. Therefore, with connecting axes at C_1, D_1 , and E_1 , a pentagon is formed. According to the geometry and link motion, relevant points in base frame are

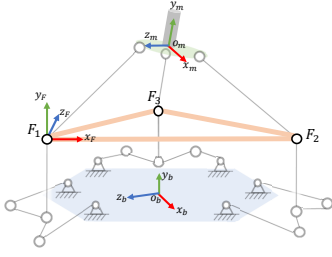
$$\mathbf{C}_1^b = [l_{AC}s\theta_1 + A_{1x}^b \quad 0 \quad l_{AC}c\theta_1 + A_{1z}^b]^T \quad (4)$$

$$\mathbf{D}_1^b = [l_{AC}s\sigma_1 + B_{1x}^b \quad 0 \quad l_{AC}c\sigma_1 + B_{1z}^b]^T \quad (5)$$

where $s(\cdot)$ and $c(\cdot)$ are the sin and cos operations respectively, σ_1 and θ_1 represent the angles of link deviating from z axis as displayed in Fig. 6b.

Then, the location of E_1 and F_1 can be calculated as

$$\mathbf{E}_1^b = \begin{bmatrix} l_{AC}s^+ + l_{AC}\tilde{u}c^- \\ 0 \\ A_{1z}^b + l_{AC}c^+ - \tilde{u}(-A_{1x}^b + l_{AC}s^-) \end{bmatrix} \quad (6)$$

Fig. 7. The frame defined by joint F .

$$\mathbf{F}_1^b = [E_{1x}^b \quad l_{EF} \quad E_{1z}^b]^T \quad (7)$$

where $s^+ = \frac{s\sigma_1 + s\theta_1}{2}$, $s^- = \frac{s\sigma_1 - s\theta_1}{2}$, $c^+ = \frac{c\sigma_1 + c\theta_1}{2}$,
 $c^- = \frac{c\sigma_1 - c\theta_1}{2}$, $\tilde{u} = \sqrt{\frac{4l_{CE}^2}{d_e^2} - 1}$ with $d_e =$
 $\sqrt{4A_{1x}^b{}^2 + 2l_{AC}^2 + 4l_{ACS}^- + 2l_{AC}^2 c^-}$.

Comparing $\mathbf{F}_1^m$ and $\mathbf{F}_1^b$ from (3) and (7), they simultaneously explicate the spatial location of the joint F_1 , although in different frames. In a similar way, F_2 and F_3 on the other two chains can also be defined in these two frames respectively. Considering the spatial distance consistency of these points in different frames, the following equations hold.

$$\begin{cases} |\mathbf{F}_1^b - \mathbf{F}_2^b| - |\mathbf{F}_1^m - \mathbf{F}_2^m| = 0 \\ |\mathbf{F}_2^b - \mathbf{F}_3^b| - |\mathbf{F}_2^m - \mathbf{F}_3^m| = 0 \\ |\mathbf{F}_3^b - \mathbf{F}_1^b| - |\mathbf{F}_3^m - \mathbf{F}_1^m| = 0 \end{cases} \implies \mathbf{F}(\mu) = \mathbf{0} \quad (8)$$

with $\mu = [\mu_1 \quad \mu_2 \quad \mu_3]$.

As known, it is difficult to solve the analytical solution for the multivariate equations. In this study, we set the multivariate equations (8) as the optimization cost function. The Newton-Raphson method is adopted for efficient iterative optimization [25]. With this numerical iteration, the unknown angles, denoted by the variables μ_1 , μ_2 and μ_3 , are attained. By a series of tests, the general error upper bound is no more than 10^{-4} rad after 4 iterations, satisfying practical demand.

3) Unit Orthogonal Spatial Vector based Orientation Recognition: The orientation of the mobile platform is usually expressed through the direction cosine matrix $\mathbf{R}_m^b \in SO(3)$. It plays a pivotal role in the derivation of kinematic equations. Considering the spatial positions of joints $F_i (i = 1, 2, 3)$, not only are the distances between them identical in different frames, but their relative locations also remain consistent. Consequently, a frame denoted by F_1 can be expressed in both the mobile platform and the base frames. The difference between the two expressions reflects the transformation from the base to the mobile platform frames, providing the desired orientation.

Define a frame with F_1 as the origin, $\overrightarrow{F_1 F_2}$ pointing to x axis, and y axis pointing upwards. As shown in Fig. 7, the unit orthogonal vectors along the corresponding axes in base frame are

$$\mathbf{v}_f^b = \begin{bmatrix} \mathbf{v}_{Fx}^b \\ \mathbf{v}_{Fy}^b \\ \mathbf{v}_{Fz}^b \end{bmatrix}^{3 \times 3} = \begin{bmatrix} \frac{\mathbf{F}_2^b - \mathbf{F}_1^b}{|\mathbf{F}_2^b - \mathbf{F}_1^b|} \\ \frac{\mathbf{v}_{Fx}^b \times (\mathbf{F}_3^b - \mathbf{F}_1^b)}{|\mathbf{F}_3^b - \mathbf{F}_1^b|} \\ o_F^b x_F^b \times o_F^b y_F^b \end{bmatrix}^{3 \times 3}. \quad (9)$$

Similarly, the same vectors expressed in mobile platform frame are

$$\mathbf{v}_f^m = \begin{bmatrix} \mathbf{v}_{Fx}^m \\ \mathbf{v}_{Fy}^m \\ \mathbf{v}_{Fz}^m \end{bmatrix}^{3 \times 3} = \begin{bmatrix} \frac{\mathbf{F}_2^m - \mathbf{F}_1^m}{|\mathbf{F}_2^m - \mathbf{F}_1^m|} \\ \frac{\mathbf{v}_{Fx}^m \times (\mathbf{F}_3^m - \mathbf{F}_1^m)}{|\mathbf{F}_3^m - \mathbf{F}_1^m|} \\ o_F^m x_F^m \times o_F^m y_F^m \end{bmatrix}^{3 \times 3}. \quad (10)$$

According to $\mathbf{R}_m^b$ in (1), a random vector p in above F_1 frame can be expressed in base frame as

$$p^b = (\mathbf{v}_f^b)^T p^f = \mathbf{R}_m^b (\mathbf{v}_f^m)^T p^f. \quad (11)$$

Since all the vectors in $\mathbf{v}_f^m$ are unit orthogonal, the direction cosine matrix $\mathbf{R}_m^b$ is

$$\mathbf{R}_m^b = (\mathbf{v}_f^b)^T \mathbf{v}_f^m. \quad (12)$$

Until now, the orientation $\mathbf{R}_m^b$ is obtained.

Considering the relative shifts among the base, mobile platform, and F_1 frames, the shift from base to mobile platform frames is

$$\mathbf{o}_m^b = \mathbf{F}_1^b - \mathbf{R}_m^b \mathbf{F}_1^m. \quad (13)$$

Taking (12) and (13) into (1), the pose of the mobile platform is

$$\mathbf{T}_m^b = \begin{bmatrix} (\mathbf{v}_f^b)^T \mathbf{v}_f^m & \mathbf{F}_1^b - \mathbf{R}_m^b \mathbf{F}_1^m \\ \mathbf{0}_{1 \times 3} & 1 \end{bmatrix}. \quad (14)$$

The forward kinematics of the haptic robot are derived. This is a hybrid method that considers both numerical optimisation and geometry motion. The optimisation iteration extends the general application of the parallel mechanical kinematic calculation. Meanwhile, the geometric derivation by spatial vector minimises the amount of unknown parameters, significantly facilitating the calculation efficiency.

B. Motion Jacobian

In addition to the kinematic mapping from the encoder values to the handle pose, the force mapping should also be presented from the MRF clutch twin outputs to the handle, necessitating the Jacobian matrix for accurate force generation. There are some commonly used methods to derive the Jacobian matrix [26], [27], the most intuitive is the differential of the forward kinematic equation. This method takes the analytic expression of forward kinematics as the prerequisite. However, since the numeric iteration in forward kinematics of this MRF robot, this method is inapplicable. The other method involves an approximating numeric approach, simulating a perturbation on each axis. The relationship between the pose variation and perturbation is recorded as the Jacobian. Nevertheless, as this method involves approximation, it reduces the accuracy of the force feedback.

In this study, to ensure the accuracy of the Jacobian matrix, we propose the structure decomposition and reintegration method, and integrate it into the reciprocal screw theory for this parallel robot Jacobian calculation [28], [29].

1) *Screw and Instantaneous Motion Mapping*: A pair of vectors define a unit screw as

$$\hat{\$} = \begin{bmatrix} \mathbf{s} \\ \mathbf{s}_o \times \mathbf{s} + \lambda \mathbf{s} \end{bmatrix} = [s_1 \ s_2 \ s_3 \ s_4 \ s_5 \ s_6]^T \quad (15)$$

where $\mathbf{s}$ is the unit vector along the screw axis, $\mathbf{s}_o$ is the location of a point on the screw axis, and λ represents the pitch length. Given a screw $\$ _r = [s_{r1} \ s_{r2} \ s_{r3} \ s_{r4} \ s_{r5} \ s_{r6}]^T$, its corresponding transpose is $\$ _r^T = [s_{r4} \ s_{r5} \ s_{r6} \ s_{r1} \ s_{r2} \ s_{r3}]$. If they satisfy

$$\$ _r^T \$ = s_{r4}s_1 + s_{r5}s_2 + s_{r6}s_3 + s_{r1}s_4 + s_{r2}s_5 + s_{r3}s_6 = 0, \quad (16)$$

these two screws are reciprocal.

A revolute joint is characterised by a unit screw, which is a screw of zero pitch that points along the joint axis. The reciprocal screws form a five-system, with the particularity that those zero-pitch reciprocal screws lie on all planes containing the axis of the revolute joint. The unit screw associated with a prismatic joint is a screw of infinite pitch aligned with the direction of the joint axis. The reciprocal screws form a five-system, with zero-pitch reciprocal screws lying on all planes perpendicular to the axis of the prismatic joint.

Multiple chains in the parallel robot are employed to establish the connection between the base and the mobile platform. Each chain is composed of links and joints. The instantaneous motion of the mobile platform can be expressed through the integration of the instantaneous motion of each joint, calculated by the screw method. Therefore, the instantaneous motion of mobile platform is

$$\$ _p = \sum_{j=1}^l \dot{q}_{j,i} \hat{\$}_{j,i} \quad \text{for } i = 1, 2, \dots, m \quad (17)$$

where $\dot{q}_{j,i}$ and $\hat{\$}_{j,i}$ represent the motion rate and unit screw of j th joint in i th chain.

Since multiple inaccessible passive joint velocities are included in (17), the instantaneous motion of the platform cannot be solved directly.

To eliminate these unknown passive joint velocities, we employ reciprocal screws. For a given kinematic chain containing g active (actuated) joints, we can identify g reciprocal screws, each being a screw that produces zero virtual work when paired with any passive joint screw in that chain. These g reciprocal screws are arranged into a matrix

$$J_{r,i} = \begin{bmatrix} \hat{\$}_{r1,i}^T & \hat{\$}_{r2,i}^T & \cdots & \hat{\$}_{rg,i}^T \end{bmatrix}^T.$$

Multiplying both sides of (17) by $J_{r,i}$ through taking the orthogonal product removes all terms containing passive joint velocities, because each reciprocal screw is orthogonal to every passive joint screw. The result is

$$J_{r,i} \$ _p = J_{q,i} \dot{\mathbf{q}} \quad (18)$$

where

$$J_{q,i} = \begin{bmatrix} \hat{\$}_{r1,i}^T \hat{\$}_{1,i} & \hat{\$}_{r1,i}^T \hat{\$}_{2,i} & \cdots & \hat{\$}_{r1,i}^T \hat{\$}_{g,i} \\ \hat{\$}_{r2,i}^T \hat{\$}_{1,i} & \hat{\$}_{r2,i}^T \hat{\$}_{2,i} & \cdots & \hat{\$}_{r2,i}^T \hat{\$}_{g,i} \\ \vdots & \vdots & \ddots & \vdots \\ \hat{\$}_{rg,i}^T \hat{\$}_{1,i} & \hat{\$}_{rg,i}^T \hat{\$}_{2,i} & \cdots & \hat{\$}_{rg,i}^T \hat{\$}_{g,i} \end{bmatrix}$$

The original equation (17) involved j unknown joint velocities (both active and passive). After multiplication by $J_{r,i}$, only the active joint velocities $\dot{\mathbf{q}}$ remain because the passive terms vanish due to reciprocity. Thus, the dimension of the system reduces from j to g . By applying this process to all kinematic chains and assembling the results, we obtain a complete set of equations describing the instantaneous motion of the mobile platform without needing to measure or compute any passive joint velocities.

2) *Robot Jacobian Matrix*: The developed haptic robot has a parallel mechanism consisting of three identical kinematic chains. Each kinematic chain, however, is not a simple serial kinematic chain. For example, the first kinematic chain contains a four-link sub-structure connecting the base to point E_1 , forming a small internal parallel topology within the kinematic chain itself.

This internal parallelism poses a problem for the reciprocal screw method described above. The method assumes that within each kinematic chain, we can clearly distinguish between active joints (actuated) and passive joints (unactuated but with unknown velocities). In a purely serial kinematic chain, this distinction is straightforward. But in a kinematic chain that itself contains parallel links, the relationship between joint velocities and the platform motion becomes more complex: some joints may be neither purely active nor purely passive with respect to the kinematic chain's overall constraint. Consequently, the identification of reciprocal screws is no longer direct, and the simple dimension reduction from j to g may fail.

Therefore, to derive the overall robot Jacobian matrix for this haptic device, we modified the reciprocal screw approach by breaking each kinematic chain into smaller serial sub-chains. The following section presents our chosen solution to this challenge.

In order to surmount the challenge posed by non-series configuration, decomposition and reintegration method is employed for the implementation of screw theory. The chain is decomposed and analysed by two parts separately, namely the four-link part and the platform part, as displayed in Fig. 8.

a) *Four-link Part Jacobian Matrix*: Regarding four-link part, the motion mapping between the MRF clutch twin axes and planar position of E_i is analysed. Take the first kinematic chain for an example in bottom right figure of Fig. 8. The coordinate values of A_1 , B_1 , C_1 , and D_1 can be obtained through geometric deduction. The corresponding screws are

$$\begin{aligned} \hat{\$}_{A,1} &= \left[\frac{s_{A,1}}{E_1 A_1 \times s_{A,1}} \right], & \hat{\$}_{C,1} &= \left[\frac{s_{C,1}}{E_1 C_1 \times s_{C,1}} \right], \\ \hat{\$}_{B,1} &= \left[\frac{s_{B,1}}{E_1 B_1 \times s_{B,1}} \right], & \hat{\$}_{D,1} &= \left[\frac{s_{D,1}}{E_1 D_1 \times s_{D,1}} \right] \end{aligned}$$

with $s_{A,1} = s_{C,1} = s_{B,1} = s_{D,1} = [0, 1, 0]^T$.

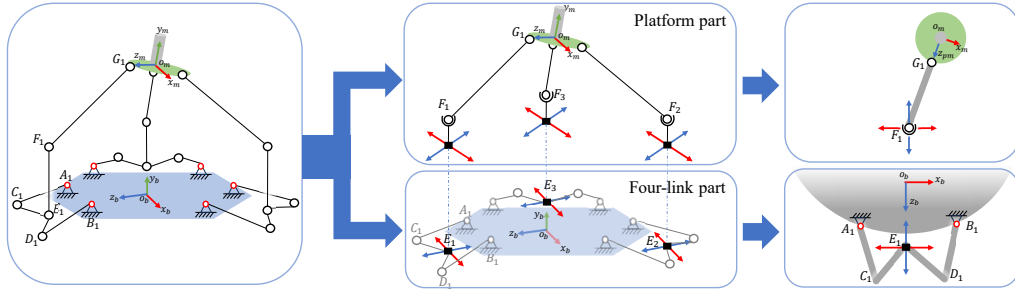


Fig. 8. Structure decomposition of the haptic robot. The robot is divided into four-link part and platform part.

Then, the screw of E_1 is

$$\begin{cases} \$_{E,1} = \dot{\theta}_1 \hat{\$}_{A,1} + \frac{\partial \angle A_1 C_1 E_1}{\partial t} \hat{\$}_{C,1} \\ \$_{E,1} = \dot{\sigma}_1 \hat{\$}_{C,1} + \frac{\partial \angle B_1 D_1 E_1}{\partial t} \hat{\$}_{D,1} \end{cases} \quad (19)$$

Set $\overrightarrow{E_1 C_1}$ and $\overrightarrow{E_1 D_1}$ as the rotation vectors of passive joints C_1 and D_1 respectively. Then, their unit reciprocal screws are

$$\begin{cases} \hat{\$}_{rC,1} = \left[\frac{\overrightarrow{E_1 C_1} / |\overrightarrow{E_1 C_1}|}{\overrightarrow{E_1 C_1} \times \overrightarrow{E_1 D_1} / |\overrightarrow{E_1 D_1}|} \right] \\ \hat{\$}_{rD,1} = \left[\frac{\overrightarrow{E_1 D_1} / |\overrightarrow{E_1 D_1}|}{\overrightarrow{E_1 C_1} \times \overrightarrow{E_1 D_1} / |\overrightarrow{E_1 D_1}|} \right] \end{cases} \quad (20)$$

Multiply $\hat{\$}_{rC,1}^T$ and $\hat{\$}_{rD,1}^T$ to (19), following equation holds.

$$\begin{bmatrix} \overrightarrow{E_1 C_1}_x & \overrightarrow{E_1 C_1}_z \\ \overrightarrow{E_1 D_1}_x & \overrightarrow{E_1 D_1}_z \end{bmatrix} \begin{bmatrix} \dot{E}_{1x} \\ \dot{E}_{1z} \end{bmatrix} = \begin{bmatrix} u_A & 0 \\ 0 & u_B \end{bmatrix} \begin{bmatrix} \dot{\theta}_1 \\ \dot{\sigma}_1 \end{bmatrix} \quad (21)$$

where $u_A = -\overrightarrow{E_1 C_1}_x \overrightarrow{E_1 A_1}_z + \overrightarrow{E_1 A_1}_x \overrightarrow{E_1 C_1}_z$ and $u_B = -\overrightarrow{E_1 D_1}_x \overrightarrow{E_1 B_1}_z + \overrightarrow{E_1 B_1}_x \overrightarrow{E_1 D_1}_z$, and the subscripts denote the results on corresponding axes and kinematic chain of the vectors or coordinate values.

Therefore, the Jacobian matrix of this four-link in first kinematic chain is

$$J_{b1} = \begin{bmatrix} \overrightarrow{E_1 C_1}_x & \overrightarrow{E_1 C_1}_z \\ \overrightarrow{E_1 D_1}_x & \overrightarrow{E_1 D_1}_z \end{bmatrix}^{-1} \begin{bmatrix} u_A & 0 \\ 0 & u_B \end{bmatrix} \quad (22)$$

Considering the vertical configuration of E_1 and F_1 and the angle interval of all kinematic chains, the four-link part Jacobian matrix from six MRF clutch twins to three F points is expressed as

$$\dot{q}_F = \begin{bmatrix} J_{b1} & \mathbf{0} \\ \mathbf{0} & J_{b3} \end{bmatrix} \dot{q}_p \quad (23)$$

where $\dot{q}_p = [\dot{\theta}_1 \ \dot{\sigma}_1 \ \dot{\theta}_2 \ \dot{\sigma}_2 \ \dot{\theta}_3 \ \dot{\sigma}_3]$ reflects the rotational rates of six MRF clutch twins, and $\dot{q}_F = [\dot{F}_{1x} \ \dot{F}_{1z} \ \dot{F}_{2x} \ \dot{F}_{2z} \ \dot{F}_{3x} \ \dot{F}_{3z}]$ denotes the planar velocity of three F points.

b) Platform Link Jacobian Matrix: Since three F points move in the horizontal plane, the kinematics of this part is equivalent to a terse structure as top figure of Fig. 8. The first kinematic chain of this parallel mechanism $F_1 G_1$ is in top right of Fig. 8. This kinematic chain has six DoFs, including two translational, three revolved DoFs at F_1 , and one revolved DoF at G_1 . The corresponding screws are

$$\begin{aligned} \hat{\$}_{1,1} &= \begin{bmatrix} \mathbf{0} \\ \mathbf{s}_1 \end{bmatrix}, & \hat{\$}_{2,1} &= \begin{bmatrix} \mathbf{0} \\ \mathbf{s}_2 \end{bmatrix}, \\ \hat{\$}_{3,1} &= \begin{bmatrix} \mathbf{s}_3 \\ \overrightarrow{o_m F_1} \times \mathbf{s}_3 \end{bmatrix}, & \hat{\$}_{4,1} &= \begin{bmatrix} \mathbf{s}_4 \\ \overrightarrow{o_m F_1} \times \mathbf{s}_4 \end{bmatrix}, \\ \hat{\$}_{5,1} &= \begin{bmatrix} \mathbf{s}_5 \\ \overrightarrow{o_m F_1} \times \mathbf{s}_5 \end{bmatrix}, & \hat{\$}_{6,1} &= \begin{bmatrix} \mathbf{s}_6 \\ \overrightarrow{o_m G_1} \times \mathbf{s}_6 \end{bmatrix}, \end{aligned} \quad (24)$$

with $\mathbf{s}_1 = \mathbf{s}_3 = [1, 0, 0]^T$, $\mathbf{s}_2 = [0, 0, 1]^T$, $\mathbf{s}_5 = \overrightarrow{F_1 G_1} / |F_1 G_1|$, $\mathbf{s}_4 = \mathbf{s}_3 \times \mathbf{s}_5 / |\mathbf{s}_3 \times \mathbf{s}_5|$. Since $F_1 G_1$ connects the horizontal plane and the mobile platform, the instantaneous motion of mobile platform frame is expressed as

$$\$m = \dot{F}_{1x} \hat{\$}_{1,1} + \dot{F}_{1z} \hat{\$}_{2,1} + \dot{\theta}_{3,1} \hat{\$}_{3,1} + \dots + \dot{\theta}_{6,1} \hat{\$}_{6,1} \quad (25)$$

In this equivalent kinematic chain, $\dot{F}_{1x}$ and $\dot{F}_{1z}$ are actively actuated translational joint rates. Other revolved joint rates are passively actuated. Because the passive joints can be deemed as the combination of a sphere joint and a revolved joint, potential screws that are reciprocal to all passive joints are located on the plane defined by F_1 and G_1 . These reciprocal screws are perpendicular to $\mathbf{n}_{\perp 1}$,

$$\mathbf{n}_{\perp 1} = [n_{x,\perp 1}, n_{y,\perp 1}, n_{z,\perp 1}]^T = \mathbf{s}_5 \times \mathbf{s}_6. \quad (26)$$

Hence, the screw that reciprocal to all joints except $\hat{\$}_{1,1}$ is

$$\hat{\$}_{r1,1} = \begin{bmatrix} \mathbf{s}_{r1,1} \\ \overrightarrow{o_m F_1} \times \mathbf{s}_{r1,1} \end{bmatrix} \quad (27)$$

where $\mathbf{s}_{r1,1} = \frac{\mathbf{n}_{\perp 1} \times \mathbf{s}_2}{|\mathbf{n}_{\perp 1} \times \mathbf{s}_2|} = \frac{[n_{y,\perp 1} \ -n_{x,\perp 1} \ 0]^T}{\tilde{n}_z}$ with $\tilde{n}_p = \sqrt{1 - n_{p,\perp 1}^2}$, $p \in \{x, y, z\}$. Similarly, the screw that reciprocal to all joints except $\hat{\$}_{2,1}$ is

$$\hat{\$}_{r2,1} = \begin{bmatrix} \mathbf{s}_{r2,1} \\ \overrightarrow{o_m F_1} \times \mathbf{s}_{r2,1} \end{bmatrix} \quad (28)$$

$$\text{with } \mathbf{s}_{r2,1} = \frac{\mathbf{n}_{\perp 1} \times \mathbf{s}_1}{|\mathbf{n}_{\perp 1} \times \mathbf{s}_1|} = \frac{[0 \ n_{z,\perp 1} \ n_{y,\perp 1}]^T}{\tilde{n}_x}.$$

Product (27) and (28) with (25), and we get

$$\begin{bmatrix} (\overrightarrow{o_m F_1} \times \mathbf{s}_{r1,1})^T & \mathbf{s}_{r1,1}^T \\ (\overrightarrow{o_m F_1} \times \mathbf{s}_{r2,1})^T & \mathbf{s}_{r2,1}^T \end{bmatrix} \$m = \begin{bmatrix} \frac{n_{y,\perp 1}}{\bar{n}_z} & 0 \\ 0 & \frac{n_{y,\perp 1}}{\bar{n}_x} \end{bmatrix} \begin{bmatrix} \dot{F}_{1x} \\ \dot{F}_{1z} \end{bmatrix} \quad (29)$$

Above equation (29) reflects the instantaneous motion mapping from F_1 to mobile platform. Then, the mapping from $\dot{q}_F$ to mobile platform is

$$J_r^\top \$m = J_F \dot{q}_p \quad (30)$$

with $\$m = [\omega_{x,m} \ \omega_{y,m} \ \omega_{z,m} \ v_{x,m} \ v_{y,m} \ v_{z,m}]^T$ denoting the motion rates of mobile platform,

$$J_r^\top = \begin{bmatrix} (\overrightarrow{o_m F_1} \times \mathbf{s}_{r1,1})^T & \mathbf{s}_{r1,1}^T \\ (\overrightarrow{o_m F_1} \times \mathbf{s}_{r2,1})^T & \mathbf{s}_{r2,1}^T \\ (\overrightarrow{o_m F_2} \times \mathbf{s}_{r1,2})^T & \mathbf{s}_{r1,2}^T \\ (\overrightarrow{o_m F_2} \times \mathbf{s}_{r2,2})^T & \mathbf{s}_{r2,2}^T \\ (\overrightarrow{o_m F_3} \times \mathbf{s}_{r1,3})^T & \mathbf{s}_{r1,3}^T \\ (\overrightarrow{o_m F_3} \times \mathbf{s}_{r2,3})^T & \mathbf{s}_{r2,3}^T \end{bmatrix}, \quad (31)$$

and

$$J_F = \text{diag}(\bar{n}(1, z), \bar{n}(1, x), \bar{n}(2, z), \bar{n}(2, x), \bar{n}(3, z), \bar{n}(3, x)) \quad (32)$$

where $\bar{n}(k, \delta) = \frac{n_{y,\perp k}}{\sqrt{1-n_{\delta,\perp k}^2}}$.

Combining (23) and (30 ~ 32), the overall Jacobian matrix from MRF clutch twin angles to mobile platform pose is

$$J = (J_r^\top)^{-1} J_F \begin{bmatrix} J_{b1} & \mathbf{0} \\ J_{b2} & \\ \mathbf{0} & J_{b3} \end{bmatrix}^{6 \times 6}. \quad (33)$$

and the instantaneous motion mapping is

$$\$m = J \dot{q}_p. \quad (34)$$

By virtue of the combination of screw theory with structural decomposition and reintegration, the Jacobian matrix of this parallel robot is obtained. The designed decomposition and reintegration method eliminates the barrier of parallel structure in each kinematic chain. It turns the intractable structure to equivalent but solvable one. The, the screw theory provides efficient and accurate Jacobian calculation solution for the parallel robot.

C. Force Jacobian

According to the effort consistency of the input and output, there is a transpose relationship between the velocity and force Jacobian matrix. Consequently, the force mapping from the mobile platform handle to the MRF clutch twins is

$$F_p = J^T F_m \quad (35)$$

where $F_p \in R^{6 \times 1}$ are the forces(torques) at the mobile platform handle, and $F_m \in R^{6 \times 1}$ are the torques on the MRF clutch twins.

D. Teleoperation control system

A teleoperation system was integrated with this parallel robot. The detailed control system workflow of this teleoperation is shown in Fig. 9. In the forward process, the mobile platform handle pose at leader side is obtained through numerical iteration based joint recognition and unit orthogonal spatial vector calculation. Then, this pose acts as command to the tool at the end-effector of the follower robot. In the feedback process, the first feedback is the visual feedback that provides explicit executing scene at following side to the operator. The second feedback is the force feedback that regenerates the forces and torques applied to the tool. In this process, the Jacobian matrix is derived according to the encoder values. Then, the force sensor measurements are mapped to the MRF clutch twins. Finally, explicit current commands are calculated for the each MRF clutch.

Single MRF clutch torque rotating at constant velocity $\dot{q}_{sc}$ is usually presented by $T_{c,s} = \tau_s(B) + \text{sign}(\dot{q}_{sc}) * f_d + \dot{q}_{sc} * f_v$, where $\tau_s(B)$ is the yield torque arising from magnetic flux density B , f_d and f_v are dead band friction and viscous coefficient. Considering the counteraction of dead band friction from two MRF clutch rotating along opposite direction, the overall torque output of this MRF clutch twin is

$$T_c = \tau(B(I_a)) - \tau(B(I_b)) + 2 * \dot{q} * f_v. \quad (36)$$

which conforms well to the viscous yield stress feature of prior study [30].

The relationship between yield stress τ and magnetic flux density B is presented by a polynomial model, and the derivation from current I to B has been expressed by a modified Bouc-Wen model [16], [31].

The motor runs at a constant speed while the current amplitudes of paired clutches modulate the torque of each MRF clutch twin output axis. The MPC method is developed to calculate and apply the current command, providing a fast response and improved accuracy [16]. The cost function at time k is

$$J_c = \sum_{i=1}^{n_p} [T_t(k+i) - T_c(k+i)]^2 + \sum_{j=1}^{m_c} \lambda [\Delta I(k+j-1)]^2 \quad (37)$$

where n_p represents horizon of prediction, m_c is the control horizon, $T_t(\cdot)$ is the desired torque, $\Delta I(\cdot)$ is the current incremental sequences, and λ is the adjusting coefficient.

In vector form, the cost function is

$$J_c = (\mathbf{T}_t - \hat{\mathbf{T}}_c)^T (\mathbf{T}_t - \hat{\mathbf{T}}_c) + \lambda (\Delta \mathbf{I})^T (\Delta \mathbf{I}) \quad (38)$$

Hysteresis and nonlinearity are inherently challenging to model accurately in MRF clutches, particularly due to frequency-dependent magnetic hysteresis [32]. To address this challenge, we adopt a two-step modeling approach. First, a polynomial model is used to describe the steady-state torque response of the MRF clutch to input current. The instantaneous linearized relationship between current variation and torque variation is represented by $\mathbf{A}_d$, calculated by differentiating the torque-current polynomial. This provides a local linear approximation for control. Second, to capture the frequency-dependent hysteresis phenomena, we adopt the modified Bouc-Wen hysteresis model. The Bouc-Wen model is chosen for

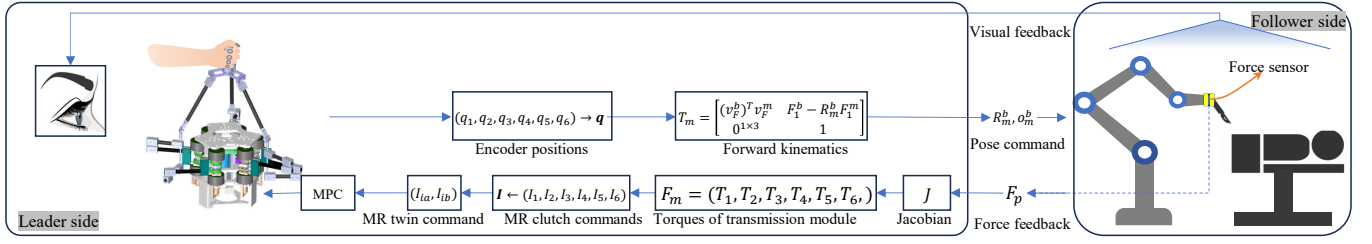


Fig. 9. The overall control loop of the teleoperation system.

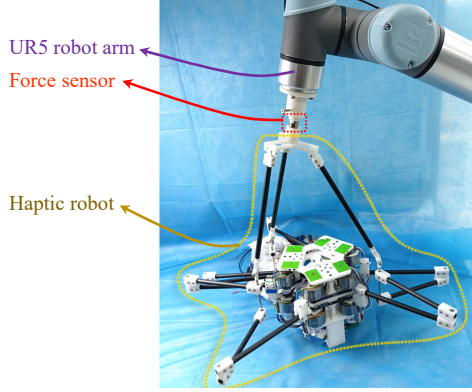


Fig. 10. Experimental setup in measuring position, orientation, force and torque accuracies.

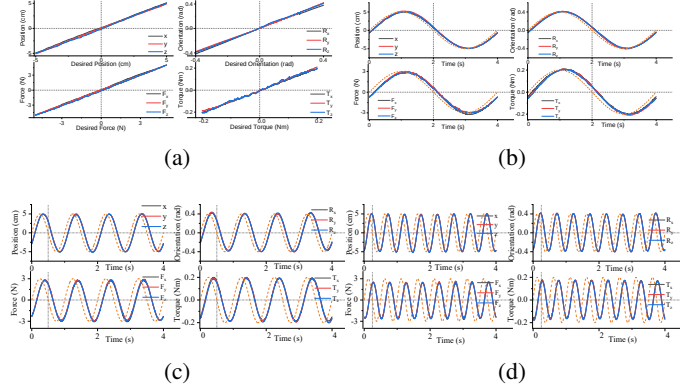


Fig. 11. Experimental results of position, orientation, force and torque accuracies of the MRF haptic robot.

its balance between computational efficiency and accuracy, as well as its widespread validation in MRF clutch applications. The model parameters are identified offline using experimental torque-current data collected under varying input frequencies.

The optimal future current inputs $\Delta \mathbf{I}$ are derived by minimizing the cost J_c . The cost J_c is minimal when $\frac{\partial J}{\partial \Delta \mathbf{I}} = 0$. Thus, the practical control sequence for minimal cost function is

$$\Delta \mathbf{I} = (\hat{\mathbf{A}}_d^T \cdot \hat{\mathbf{A}}_d + \lambda \mathbf{E})^{-1} \hat{\mathbf{A}}_d^T \cdot (\mathbf{T}_t - \mathbf{T}_0) \quad (39)$$

The current commands ($I_1, I_2, I_3, I_4, I_5, I_6$) are updated according to the desired force generation. In each MRF clutch twin, the choice of which clutch to activate depends on the sign of the current command (direction of desired torque). The stability of the control algorithm has been validated in prior work [33].

V. EXPERIMENT AND RESULTS

A. Force and Position Accuracy Assessment

Since force rendering and pose recognition are the crucial functions for a haptic robot, this experiment measures the accuracy of force and position. Following commonly used evaluation indicators in [34], the experimental configuration is displayed in Fig. 10. In comparison to the original haptic robot, it is notable that the handle is integrated with a force sensor (Nano17, ATI Industrial Automation, USA). The force sensor combines the haptic robot and the end-effector of a robotic arm (UR5e, Universal Robots A/S, Denmark).

The outline of this haptic is displayed in Fig. 10. While the mechanism may not appear compact compared to conventional motor-driven haptic devices, it is relatively compact among 6-DoF haptic robots that employ MRF actuators, which typically require additional space for clutch housings, fluid circulation, and magnetic components.

In this experiment, the robotic arm drove the haptic robot mobile platform, which moved along the x , y and z axes respectively at static, 0.25, 1, and 2 Hz. The pose of the mobile platform was reflected by both haptic robot encoders through the forward kinematics and the robotic arm. The pose repeatability of the robotic arm is 0.03 mm and meets the requirement in hand operations. At the same time, the haptic robot generated a corresponding force on the mobile platform in constant proportion to the displacement. The practical forces the force sensor were also measured simultaneously.

The measurements of pose and force are shown in Fig. 11. The dotted orange line is the desired value. Regarding the static pose accuracy, it shows that the maximum pose deviations of this proposed haptic robot are 2.8 mm in z direction and 22 mrad in z direction. Regarding the static force representation accuracy, the maximum deviations are 0.15 N in z direction and 18 mNm in z direction. However, with the test frequency increasing, the measured position and force present more lag and worse accuracy. At higher frequency, the robot frequency response is more dominant than instantaneous error in utilization. At 2 Hz, the position measurement shows phase lag at 1.26 rad and amplitude remains about 95%. The force and torque lag is about 1.9 rad and amplitude remains about

85%.

We compare the test results against established benchmarks. For force accuracy, Chua and Okamura [35] demonstrated that force errors below 0.15 N are acceptable for delicate tissue telemanipulation in robot-assisted minimally invasive surgery. Additionally, psychophysical studies by Khabbazi et al. [36] show that just noticeable differences for forces below 1 N are on the order of 0.1–0.2 N, indicating that our 0.15 N deviation falls near or below the human perception threshold. For pose accuracy, while neurosurgical navigation systems require sub-millimeter precision (≤ 0.5 mm) [citation], our target telesurgical application involves manipulation of soft tissues where positional errors up to 3 mm are clinically acceptable. Therefore, considering practical hand perception capacity and operational accuracy, this MRF haptic robot satisfies basic pose and force requirements for the intended telesurgical application.

B. Teleoperation Stability

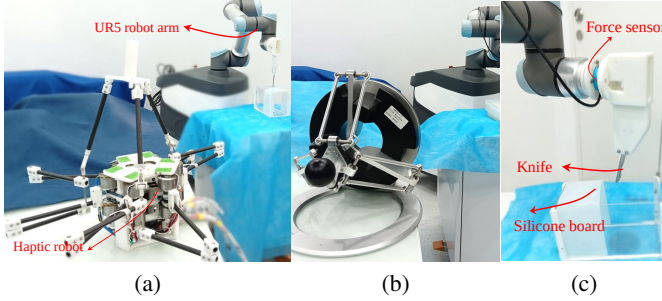


Fig. 12. Experimental configuration in teleoperation. (a) The configuration with the developed haptic robot. (b) The configuration with commercial haptic robot. (c) The silicone board and surgical tool at robotic end effector.

Stability is crucial and fundamental for a teleoperation system to perform tasks. In this experiment, the setup scene is shown in Fig. 12. Two different configurations are built at the leader side, one with the proposed haptic robot as in Fig. 12a and the other with a commercial haptic robot (omega, Force Dimension, Switzerland) as in Fig. 12b.

The comparative experiments share the same setup on the follower side. It is configured as: a UR5e robotic arm is used and a surgical knife is attached to its end-effector by a force sensor (M3703A, Sunrise Instruments, China) as in Fig. 12c. This allows the forces on the knife to be accurately measured. The target is a silicon board with 4 mm in thickness (tear strength at 25 N/mm) and the goal is to cut the board. The sampling and positioning control frequency is 50 Hz. On the leader side, gravity is compensated, and the Python code (servo gain=100 in *urx* package) runs on the same host computer. The experimental process is summarised as follows. First, the operator controlled the haptic robot to manoeuvre the knife to penetrate the silicone board vertically, and then move the knife horizontally to cut the silicone board apart. The operator perceived the regenerated force on his hand and adjusted the knife pose to avoid excessive force that could cause the knife to bend or even break. The MPC control is adopted in the proposed haptic robot force regeneration. High stiffness and

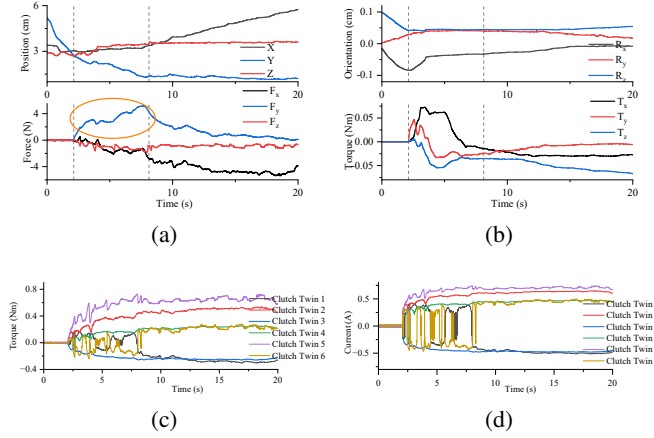


Fig. 13. Teleoperation results using the proposed MRF-based haptic robot. (a) Position and interaction force. (b) Orientation and interaction torque. (c) Output torques of the six MRF clutch twins. (d) Corresponding clutch-current commands.

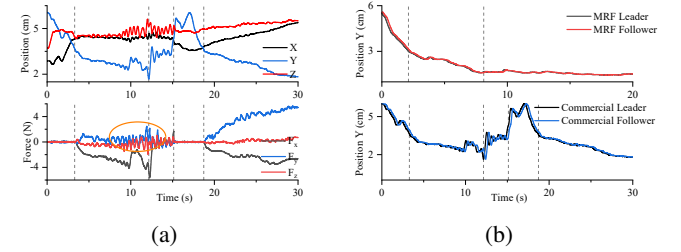


Fig. 14. Comparison of teleoperation results using the commercial haptic device and the proposed MRF-based haptic robot. (a) Position and interaction force obtained using the commercial haptic device. (b) Comparison of the y -direction position and interaction-force responses of the proposed and commercial haptic devices.

fast force response are achieved in the proposed MRF robot with customised control method.

The Fig. 13a shows the results of this teleoperation task with the proposed haptic robot. The overall task execution is smooth and stable. As in Fig. 13a and 13b, in the first 2s, the knife moves slowly downwards along the y direction. During this process, the knife has no contact with the silicone board, so no force is detected. During 2–8s the knife tries to penetrate the board. The resistance on the knife is reflected by the forces in opposite directions. The forces increase as the knife moves because of the elastic deformation of the board. As shown by the orange ellipse, some waves are observed indicating the operator's adjustment in dealing with the resistance force in this process. By 8s, the board is penetrated and the force in the penetrating y direction decreases. In the remaining time, the knife moves along x direction to cut the silicone apart. Meanwhile, the positions and forces just have minor fluctuations in other directions. Throughout the whole process, the force and position fluctuate on an even keel and only small waves are observed. The Fig. 13c and Fig. 13d show the torque and current command to each MRF clutch twin, reflecting the algorithm details of the MRF robot.

Fig. 14a shows the position and force curves in this teleoperation task with the commercial haptic robot. Similarly,

in 0-3s, the knife moves slowly without touching the silicon board. Then the direct interaction starts with the operator's intention for penetration until 12s when the silicone board is penetrated after a series of fierce vibrations. Then, in 12-15s, the operator has to relieve the tense interaction to avoid more vibrations by lifting the surgical tool. In 15-19 s, the operator resettles the knife. After that, the operator continues to steer the knife carefully, cutting the silicone sheet apart. As a typical example in the orange ellipse, the force and position curves show obvious fluctuations during the downward movement of the knife. Throughout the whole process, force and position vary with significant fluctuations and it takes the operator more time and actions to deal with them. Fig. 14b displays the position along the penetrating direction y axis in leader and follower robots. Two subfigures refer to the two conditions of MRF and commercial robots as leader respectively. Slight deviation can be found between the leader and follower curves. The MRF robot as leader has smooth manipulation and leads to good curve overlapping. However, the commercial robot as leader results in vibrations and obvious position deviation can be observed.

The proposed MRF haptic robot offers clear benefits over the commercial device (Fig. 13b). (1) Smoother operation: the proposed system exhibits minimal force/position fluctuations, whereas the other produced severe vibrations requiring task interruption; (2) Faster task completion: penetration time reduced from 19 s to 8 s (58% improvement); These advantages are particularly valuable for telesurgery, where haptic stability directly impacts tissue safety and procedure efficiency. However, because the two devices were not independently optimized according to their respective control architectures, these observations should be interpreted only as system-level results under the implemented experimental conditions.

Compared with the commercial robot, which offers a workspace of $\varnothing 160 \times 110$ mm and a maximum force of 12 N, our proposed robot provides a smaller workspace of $100 \times 100 \times 100$ mm translation and ± 0.8 rad rotation per axis, with a maximum force of 8 N. Despite of this, the proposed MRF haptic robot maintains passive damping and low reflected inertia characteristics of the clutch-based architecture. The present workspace was sufficient for the localized proof-of-concept tasks demonstrated in this study, but may require re-indexing during larger or wrist-intensive motions.

C. Discussion

This section discusses the experimental findings, the architectural contributions of the proposed haptic robot, and the limitations of the current prototype.

Force and Position Accuracy: The first experiment compared the commanded pose and force of the haptic robot with the corresponding experimental measurements. The results verified the basic position, force, and torque rendering accuracy of the proposed device under laboratory conditions. Considering the target ranges adopted for the present soft-tissue manipulation tasks and the practical sensitivity of human perception [37], [38], the prototype demonstrated proof-of-concept performance for benchtop surgical teleoperation.

However, clinical validation remains necessary before conclusions can be drawn regarding suitability for practical surgical use.

The parallel mechanism contributes to a compact structure and low reflected inertia at the operator interface, consistent with related parallel haptic designs [39]. In addition, the bidirectional MRF clutch twins enable active force and torque generation while mechanically decoupling the operator interface from direct interaction with the motor rotor.

Teleoperation Stability: The second experiment compared the implemented teleoperation systems using the commercial haptic device and the proposed MRF-based haptic robot. Under the controllers and operating conditions used in this study, the proposed MRF system exhibited less oscillatory behavior during silicone penetration and cutting. The passive damping of the MRF clutches may have contributed to this robustness.

However, the two devices were not independently optimized according to their respective control architectures. In particular, both systems shared the same 50 Hz bilateral teleoperation update rate, while the proposed MRF robot employed a customized MPC-based force controller. Therefore, the observed difference should be interpreted as a system-level result under the tested conditions rather than as evidence of a universal stability advantage of MRF actuation over appropriately tuned motor-driven haptic devices.

The teleoperation experiment is also influenced by the follower robot, controller implementation, communication and computation delays, environmental properties, and operator behavior. It should therefore be regarded as an application-level proof-of-concept demonstration rather than a substitute for standardized device-level characterization.

The present prototype also has a smaller workspace than the commercial device. It is therefore most suitable for localized surgical manipulation tasks with limited motion ranges, while its applicability to tasks requiring large translational excursions, frequent re-indexing, or repeated wrist rotations remains to be established.

Architectural Contributions: To the authors' knowledge, this is the first MRF-based haptic robot that enables active bidirectional force and torque generation in all six DoFs using a single motor and six clutch twins. The principal contribution of the work is therefore the proposed actuation and mechanical architecture, rather than general performance superiority over commercial motor-driven haptic devices.

The parallel mechanism provides a compact structural arrangement, while the clutch-based transmission maintains low reflected inertia at the operator interface. To address the forward-kinematics problem of the parallel mechanism, iterative optimization was combined with unit orthogonal spatial-vector analysis. To derive the Jacobian matrix, a structural decomposition and reintegration method was introduced and implemented using screw theory [40], [41].

The experiments demonstrate the feasibility of the proposed mechanical architecture, kinematic solution, force-generation principle, and teleoperation implementation under laboratory conditions. However, standardized characterization of Z-width, force-rendering bandwidth, maximum stable stiffness,

and force/torque resolution remains necessary before broader claims regarding haptic performance can be made.

Operational Safety: The MRF-based architecture has several characteristics that may be advantageous from a safety perspective. At zero clutch current, commanded clutch torque is removed and the operator primarily experiences the passive mechanical resistance of the transmission and MRF rather than direct motor torque. In addition, the clutch mechanism mechanically separates the operator interface from the motor rotor, which can reduce reflected motor inertia.

Nevertheless, the device should not be regarded as intrinsically fail-safe solely because the clutch currents can be reduced to zero. Practical safety also depends on the complete mechanical, electrical, sensing, and control architecture. Accordingly, further fault-mode analysis and dedicated safety validation will be required before clinical deployment.

Operating Stability: The proposed MRF architecture has several characteristics that may support stable contact interaction. First, the MRF clutches dissipate energy through fluid shear, providing passive damping during interaction with stiff environments. Second, the operator interacts primarily with the low-inertia clutch output components rather than directly with the motor rotor. Third, the clutch output torque can be regulated through the applied current.

Nevertheless, the stability of both MRF-based and motor-driven haptic systems depends on the complete system architecture, including actuator dynamics, controller design, sampling rate, communication and computation delays, device dynamics, environmental impedance, and mechanical damping.

The intrinsic response time of the MRF material or clutch should also be distinguished from the end-to-end response of the complete system. At 2 Hz, the measured force and torque phase lag was approximately 1.9 rad, corresponding to an equivalent delay of approximately 151 ms. This system-level delay may reduce the range of stiffness that can be rendered stably. Although the passive damping of the MRF clutches may contribute to contact robustness, it does not eliminate delay-induced instability. Accordingly, the present results demonstrate stable operation under the tested conditions but do not establish a large Z-width, high maximum stable stiffness, or immunity to instability during high-stiffness interaction.

Limitations and Future Work: Several limitations of the current prototype must be acknowledged.

First, MRF sedimentation remains an engineering challenge because suspended particles may settle during prolonged idle periods. The experiments in this study were conducted using freshly prepared MRF, while future work will investigate long-term fluid management strategies such as periodic recirculation and improved material formulations.

Second, thermal effects may influence MRF viscosity and clutch output characteristics. The present experiments were conducted over relatively short durations under controlled laboratory conditions. Future work will investigate temperature monitoring and compensation for prolonged operation.

Third, environmental contamination remains a concern for MRF-based devices. The present design employs a permanent magnetic-ring sealing mechanism, as shown in Fig. 1a, based

on the principle reported by Vo et al. [19]. Further long-duration leakage and durability testing will nevertheless be required.

Fourth, the current prototype generates noticeable acoustic noise from the transmission system. Future work will investigate lower-noise transmission components, vibration isolation, and enclosure design.

Fifth, the present study does not provide a complete standardized characterization of the haptic device. In particular, the Z-width, force-rendering bandwidth, maximum stable virtual stiffness, and force/torque resolution have not yet been quantified under controlled single-device test conditions. The teleoperation experiment is influenced by the follower robot, controller, communication delay, environment, and operator, and therefore cannot substitute for these measurements. Future work will systematically characterize these metrics for representative translational and rotational axes before extending the evaluation to the complete six-DoF workspace.

Finally, although the current prototype successfully completed several proof-of-concept suturing demonstrations, its limited translational and rotational workspace may require re-indexing during tasks involving larger excursions or repeated wrist rotations. Therefore, the present results do not establish that the approximately ± 0.8 rad rotational range is sufficient for the full range of wrist-intensive surgical procedures. Future work will focus on enlarging the mechanical workspace and developing motion-scaling and re-centering strategies to improve dexterity.

VI. CONCLUSION

This study presented a 6-DoF parallel haptic robot based on MRF clutches. The principal contribution is a single-motor actuation architecture that enables active bidirectional force and torque generation through six clutch twins while maintaining low reflected inertia at the operator interface. To support the implementation of the parallel mechanism, a hybrid forward-kinematics method combining iterative optimization and spatial-vector analysis was developed. In addition, the Jacobian matrix was derived using a structural decomposition and reintegration method implemented through screw theory.

The experiments demonstrated the feasibility of the proposed mechanical architecture, kinematic formulation, force-generation principle, and benchtop teleoperation implementation. Under the specific controllers and operating conditions used in this study, the proposed MRF-based system exhibited stable operation and less oscillatory behavior during the demonstrated silicone penetration and cutting task. However, these results should be interpreted as proof-of-concept validation rather than as evidence of universal superiority over commercial motor-driven haptic devices.

The current prototype remains limited by nonzero free-space resistance, acoustic noise, restricted workspace, and system-level response delay. Moreover, standardized characterization of Z-width, force-rendering bandwidth, maximum stable stiffness, and force/torque resolution remains necessary before broader conclusions regarding haptic performance can be made.

Future work will therefore focus on comprehensive device-level characterization, reduction of system delay and acoustic noise, improvement of free-space transparency, enlargement of the usable workspace, and validation in more representative surgical tasks. Overall, the proposed architecture provides a new MRF-based approach to six-DoF active haptic interaction and establishes a foundation for further development of compact, low-inertia, and passively damped haptic interfaces.

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